

1. Let $a \in \mathbb{Z}$ and $[t]$ be the greatest integer $\leq t$, then the number of points, where the function $f(x) = [a + 13 \sin x]$, $x \in (0, \pi)$ is not differentiable, is

[2023 (06 Apr Shift 1)]

2. Let $f : (-2, 2) \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} x[x] & , -2 < x < 0 \\ (x-1)[x] & , 0 \leq x < 2 \end{cases}$ where $[x]$ denotes the greatest integer function. If m and n respectively are the number of points in $(-2, 2)$ at which $y = |f(x)|$ is not continuous and not differentiable, then $m + n$ is equal to _____.

[2023 (10 Apr Shift 1)]

3. Let $f(x) = [x^2 - x] + [-x + [x]]$, where $x \in \mathbb{R}$ and $[t]$ denotes the greatest integer less than or equal to t . Then, f is

[2023 (11 Apr Shift 1)]

- (1) continuous at $x = 0$, but not continuous at $x = 1$
- (2) continuous at $x = 1$, but not continuous at $x = 0$
- (3) continuous at $x = 0$ and $x = 1$
- (4) not continuous at $x = 0$ and $x = 1$

4. Let f and g be two functions defined by $f(x) = \begin{cases} x+1, & x < 0 \\ |x-1|, & x \geq 0 \end{cases}$ and $g(x) = \begin{cases} x+1, & x < 0 \\ 1, & x \geq 0 \end{cases}$. Then $(g \circ f)(x)$ is

[2023 (11 Apr Shift 2)]

- (1) Continuous everywhere but not differentiable exactly at one point
- (2) Continuous everywhere but not differentiable at $x = 1$
- (3) Differentiable everywhere
- (4) Not continuous at $x = 1$

5. Let $[x]$ be the greatest integer $\leq x$. Then the number of points in the interval $(-2, 1)$ where the function $f(x) = [x] + \sqrt{x - [x]}$ is discontinuous, is _____.

[2023 (12 Apr Shift 1)]

6. Let $[x]$ denote the greatest integer function and $f(x) = \max\{1 + x + [x], 2 + x, x + 2[x]\}$, $0 \leq x \leq 2$, where f is not continuous and n be the number of points in $(0, 2)$, where f is not differentiable. Then $(m + n)^2 + 2$ is equal to

[2023 (15 Apr Shift 1)]

- (1) 2
- (2) 11
- (3) 6
- (4) 3

ANSWER KEYS

1. (25)

2. (4)

3. (2)

4. (1)

5. (2)

6. (4)

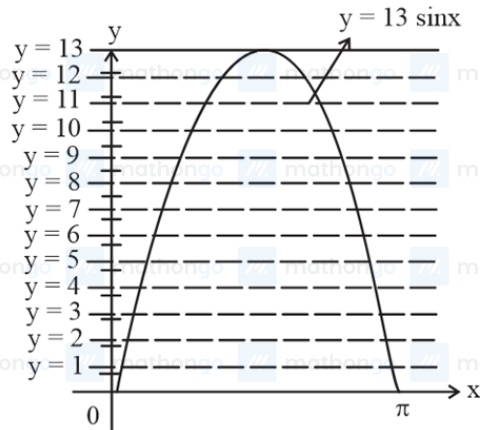
1. (25)

Given,

$$f(x) = [a + 13 \sin x], \quad x \in (0, \pi)$$

$$\Rightarrow f(x) = [13 \sin x] + a, \text{ as } a \in \text{Integer}$$

Now let $g(x) = 13 \sin x$, now plotting the diagram of $y = 13 \sin x$ we get,



Now from above diagram we can see that,

$$f(x) = [13 \sin x] \text{ is not differentiable at intersection points of } y = k, \quad k \in [1, 13] \text{ \& } k \in \mathbb{Z}$$

$$\Rightarrow \text{Points of non-differentiability} = 12 \times 2 + 1 = 25$$

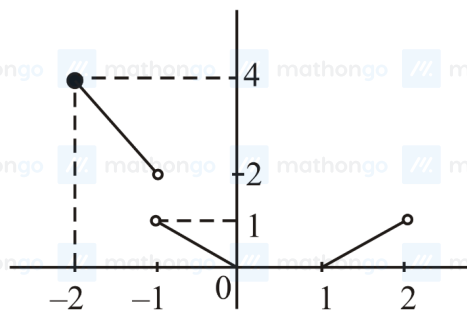
2. (4)

Given,

$$f: (-2, 2) \rightarrow \mathbb{R} \text{ be defined by } f(x) = \begin{cases} x[x], & -2 < x < 0 \\ (x-1)[x], & 0 \leq x < 2 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} -2x, & -2 < x < -1 \\ -x, & -1 \leq x < 0 \\ 0, & 0 \leq x < 1 \\ x-1, & 1 \leq x < 2 \end{cases}$$

Now plotting the diagram of the above function we get,



Now from above diagram we can say that, $y = f(x)$ is same as $y = |f(x)|$

Hence, the function is not continuous at one point and non differentiable at three points, so $m = 1, n = 3$

$$\text{Hence, } m + n = 4$$

3. (2)

Given,

$$\begin{aligned} f(x) &= [x^2 - x] + |-x + [x]| \\ \Rightarrow f(x) &= [x^2 - x] + |x - [x]|, \{ \text{as } |-A| = |A| \} \\ \Rightarrow f(x) &= [x^2 - x] + \{x\}, \{ \text{as } [x] + \{x\} = x \} \\ \Rightarrow f(x) &= [x^2 - x] + \{x\} \quad (\because \{x\} \geq 0) \end{aligned}$$

Now at $x = 0$, $f(0) = 0$

And $f(0^+) = -1$, {as $x^2 - x < 0$ for $x \rightarrow 0^+$ }

So, function is discontinuous at $x = 0$

Now at $x = 1$, $f(1) = 0$,

$f(1^+) = 0 + 0 = 0$ and $f(1^-) = -1 + 1 = 0$

Hence, the function is continuous at $x = 1$.

4. (1)

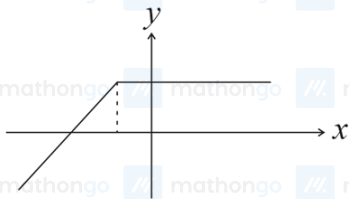
Given,

$$\begin{aligned} f(x) &= \begin{cases} x+1, & x < 0 \\ |x-1|, & x \geq 0 \end{cases} \text{ and } g(x) = \begin{cases} x+1, & x < 0 \\ 1, & x \geq 0 \end{cases} \\ \Rightarrow f(x) &= \begin{cases} x+1, & x < 0 \\ 1-x, & 0 \leq x < 1 \\ x-1, & x > 1 \end{cases} \text{ and } g(x) = \begin{cases} 1+x, & x < 0 \\ 1, & x \geq 0 \end{cases} \end{aligned}$$

Now, for $g \circ f(x)$

$$\begin{aligned} g \circ f(x) &= \begin{cases} 1 + f(x), & f(x) < 0, x < 0 \\ 1, & f(x) \geq 0, x \geq 0 \end{cases} \\ \Rightarrow g \circ f(x) &= \begin{cases} x+2, & x < -1 \\ 1, & x \geq -1 \end{cases} \end{aligned}$$

Now plotting the diagram we get,



Now from above diagram we can say that function is continuous but not differentiable at $x = -1$ due to sharp corner.

5. (2)

Given,

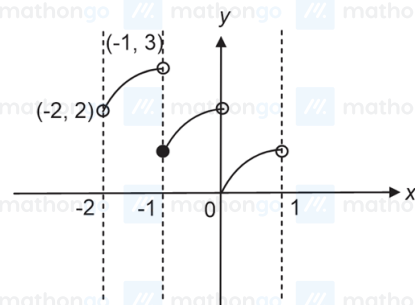
$$f(x) = |[x]| + \sqrt{x - [x]}$$

$$x - [x] \geq 0 \Rightarrow x \in \mathbb{R}$$

Now using the property of greatest integer function and simplifying the function we get,

$$f(x) = \begin{cases} 2 + \sqrt{x+2}, & -2 < x < -1 \\ 1 + \sqrt{x+1}, & -1 \leq x < 0 \\ \sqrt{x}, & 0 \leq x < 1 \end{cases}$$

Now plotting the diagram we get,



Now from the diagram we can say that, $f(x)$ is discontinuous at two points $x = \{-1, 0\}$.

6. (4)

Given,

$$f(x) = \max\{1 + x + [x], 2 + x, x + 2[x]\}, \quad 0 \leq x \leq 2$$

$$\Rightarrow f(x) = \begin{cases} \max\{x + 1, x + 2, x\} & 0 \leq x < 1 \\ \max\{x + 2, x + 2, x + 2\} & 1 \leq x < 2 \\ \max\{5, 4, 6\} & x = 2 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} x + 2, & 0 \leq x < 1 \\ x + 2, & 1 \leq x < 2 \\ 6, & x = 2 \end{cases}$$

$$\Rightarrow f(x) = \begin{cases} x + 2 & 0 \leq x < 2 \\ 6 & x = 2 \end{cases}$$

So, f is not continuous at $x = 2$ as $f(2^-) = 4$ & $f(2) = 6$

And f is differentiable in $(0, 2)$ as $x + 2$ is a linear function,

$$\therefore m = 1, n = 0$$

$$\Rightarrow (m + n)^2 + 2 = 1 + 2 = 3$$

Hence this is the required option.